

DAY 8 Task : Building Your First Agent & Database Setup

This report outlines the fundamental concepts for building a functional AI agent in n8n and the steps for setting up external databases (Postgres and Supabase) to enhance agent memory and capabilities.

Part 1: Key Concepts in Building Your First AI Agent

Building an AI agent goes beyond just connecting to a language model. A successful agent is comprised of several key components working together.

The Anatomy of an AI Agent .1

An AI agent is a system designed to understand a request, make decisions, and perform actions. The core components are:

- **Brain (Chat Model):** This is the Large Language Model (LLM) that provides reasoning and decision-making. You connect this via a "Chat Model" node (e.g., using OpenAI or Open Router to access various models).
- **Instructions (System Prompt):** This is the most critical part. It's the agent's "job description." You tell the agent *who* it is, *what* its role is, *what* its tools are, and any rules it must follow.
- **Tools:** These are the actions the agent can perform. In n8n, tools are other nodes connected to the agent (e.g., Google Calendar, Gmail, Google Sheets, API calls). The agent's "magic" is its ability to *decide which tool to use* based on the user's request.
- **Memory:** This allows the agent to recall previous parts of the conversation.
 - **Simple Memory:** Built-in n8n memory for short-term conversation history.
 - **External Memory (e.g., Postgres):** A persistent, external database to remember conversations over the long term.
- **Trigger:** The mechanism that activates the agent (e.g., a chat message, a webhook, an email).
- **User Message:** The specific, dynamic input from the user (e.g., "Send an email to Phil").

The Building Process .2

The process of building the agent involves:

1. **Setting the Trigger:** Start with a "Chat Trigger" (or another trigger) to receive user input.
2. **Adding the AI Agent Node:** This is the central hub.

- .**Connecting the Brain:** Attach a "Chat Model" node (like Open Router) to the agent .3
- .**Adding Memory:** Attach a "Memory" node (like "Simple Memory" to start) .4
- s: Connect all the action nodes the agent might need (e.g., "Google Sheets," "Gmail," .5
:"Google Calendar"). For each tool, you must
 - .Give it an intuitive name (e.g., "ContactDatabase") •
 - o it to let the model define its parameters (e.g., let the AI decide the "subject" and •
."body" of an email)
- .**Writing the System Prompt:** This is an iterative process .6
 - .Start with a simple prompt (e.g., "You are a helpful assistant.") •
 - he agent. When it fails or does something wrong, *reactively* update the prompt to •
.correct it
 - the agent fails to find an email before sending a message, you add an instruction: •
must always look in the `ContactDatabase` tool *before* using the `SendEmail` tool

Part 2: Postgres & Supabase Setup

- your agent persistent, long-term memory and knowledge, you need an external database.
- o a service that provides both a standard Postgres database (great for chat memory) and a
.vector store (great for RAG/knowledge)

Prerequisite: Create a Supabase Project .1

- .Sign up for a free Supabase account .1
- ".Create a new "Project .2
- oject creation, you will be asked to create a **database password**. This is critical. **Copy** .3
.and save this password immediately. You will need it for the Postgres connection

Connecting Postgres (for Chat Memory) .2

- .This database will store the conversation history
- .In n8n, add the **Postgres Chat Memory** node to your AI agent .1
- ".Click "Credential" and select "Create New .2
- Supabase project, go to the **Connect** button on the dashboard (or Project Settings > .3
.Database)
- .Select **Postgres SQL** and **Transaction pooler** .4

u will see a connection string. Find and copy the following details from it into your n8n .5
:credentials

Host: The URL (e.g., `aws-0-region.pool.supabase.com`) •

Database: `postgres` •

User: The username (e.g., `postgres.[project_id]`) •

Port: `6543` •

Password: The database password you saved when creating the project •

dential in n8n. It should test successfully. This agent will now remember conversations .6
.permanently in a table called `n8n_chat_histories` in your Supabase project

Connecting Supabase (for Vector Store / RAG) .3

tabase will store your long-term knowledge (e.g., PDFs, documents) for the agent to query

.In n8n, add a **Supabase Vector Store** node as a *tool* for your agent .1

".Click "Credential" and select "Create New" .2

.In your Supabase project, go to **Project Settings > API** .3

:Find and copy the following details into your n8n credentials .4

Host: Copy the `Project URL` •

Service Role Secret: Copy the `service_role` secret key (it's hidden by default) •

.Save the credential in n8n .5

Critical Final Step: Initialize the Database

base Vector Store node in n8n needs a specific table and function in your database to .1
.work

.In your Supabase project, go to the **SQL Editor** (the "SQL" icon in the left menu) .2

.Click **New query** .3

Initialization SQL command from the n8n or Supabase documentation (it's a script that .4
.creates a table named `documents` and a function named `match_documents`)

.Paste this entire SQL command into the query window and click **RUN** .5

; script succeeds, your Supabase Vector Store is ready to have documents added to it .6
.and be used by your agent for RAG